Hamza Aziz

AI Product Engineer @VoicePlug Inc.

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PROFESSIONAL SUMMARY

AI Engineer with **3 years of experience** in **Computer Vision, Conversational/Generative AI, full-stack LLM applications, and applied MLOps**, delivering production-grade AI systems across education and enterprise environments. Built and deployed **multiple facial recognition platforms** that reduced manual operational effort by **up to 90%**, achieving **>99.99% accuracy** through robust real-time pipelines and automated **Milvus vector database** updates. Experienced in developing **scalable conversational AI systems**, including agentic, real-time voice ordering agents built with **LangGraph and Google Gemini**, alongside earlier Rasa-based voice assistants, with a focus on reliability, accuracy, and real-world usability. Comfortable shipping **full-stack LLM applications end-to-end** – FastAPI + Next.js services with **retrieval-augmented, citation-grounded narrators** over vLLM / OpenAI-compatible endpoints. Strong hands-on expertise in **GPU-accelerated inference** (TensorRT, Triton, DeepStream) and **production deployments** using CI/CD workflows.

SKILLS

- **Programming Languages:** Python, Bash, SQL
- **AI/ML Frameworks:** TensorFlow, PyTorch, InsightFace, YOLO, Scikit-learn, OpenCV, NumPy, Pandas
- **Generative & Conversational AI:** LangChain, LangGraph, Agentic AI (ReAct), Function Calling / Tool Use, Rasa, Gemini 2.5 / 2.0, Large Language Models (LLMs), NLP, Prompt Engineering, Retrieval-Augmented Generation (RAG), vLLM
- **Model Optimization & Inference:** TensorRT, ONNX, DeepStream SDK
  Backend & Web Development: FastAPI, python-socketio, Pydantic, Jinja2, Next.js 14, React, TypeScript, Tailwind CSS, REST APIs
- **Deployment & MLOps:** CI/CD Pipelines, Data Pipeline Automation, Docker, NVIDIA Jetson (Nano, Orin Nano), DGX A100, RTX 4090, Linux, Git
- **Databases & Storage:** Milvus (Vector DB), PostgreSQL, MongoDB, Redis, MinIO / S3-compatible Object Storage
- **Streaming & Messaging Systems:** Confluent Kafka, Flask-SocketIO
- **Other Tools:** SpeechRecognition, Tesseract OCR, Neo4j, camelot (PDF tables), cloudscraper, VSCode, PyCharm, Postman

WORK EXPERIENCE

AI Product Engineer *VoicePlug Inc.* | September 2025 – Present

- Working on a production-grade DriveThru voice ordering platform serving multiple QSR brands across live deployments.
- Contributed to the modernization from a Rasa-based NLU platform to an agentic LangGraph + Google Gemini stack, working on entity extraction, conversation flow, and tool-use orchestration.
- Helped improve client success metrics from ~25–30% to 60%+ by addressing intent recognition, entity ambiguity, and menu understanding issues.
- Supported stabilization of live deployments by improving error handling, logging, fallback escalation, and per-restaurant prompt templating across POS integrations.
- Worked closely with cross-functional teams (product, support, QA) to iterate on conversational quality based on real user interactions.

Assistant Engineer – AI *Global InfoVentures Pvt. Ltd.* | August 2023 – August 2025

- Led development of 4 commercial-grade Facial Recognition Attendance Systems on DGX A100, RTX 4090, Jetson Orin Nano, achieving >99.99% accuracy.
- Developed G6 Voice Assistant, converting natural language voice queries into precise SQL queries via Rasa, Gemini 2.0, and Milvus Vector DB; handled ambiguous queries with advanced intent disambiguation, achieving near-100% intent classification accuracy.

- Mitigated domain shift in facial recognition using a CI/CD-style pipeline combining mobile and CCTV datasets — improved cross-domain recognition by 14%.
- Automated nightly face vector updates from 8ft-high CCTV, boosting real-world accuracy from 85% to 99%.
- Integrated PostgreSQL for recognition logs and built an API sync module with legacy SIM ERP Portal, enabling real-time attendance updates and SMS alerts for over 2,000+ students, enhancing parent engagement and system transparency.

AI Intern *Global InfoVentures Pvt. Ltd.* | February 2023 – July 2023

- Conducted 10+ hands-on AI/ML training sessions and mentored 50+ students across CSE, AIML, and Data Science domains.
- Guided 10+ object detection projects using YOLOv8 and CNNs (VGG, ResNet, InceptionNet), resulting in successful full-cycle model deployments.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani – WILP 2025 – 2027
Master of Technology (Executive)

- Artificial Intelligence and Machine Learning

Dr. A. P. J. Abdul Kalam Technical University, ABES Institute of Technology
Bachelor of Technology 2019 – 2023

- Computer Science and Engineering

PROJECTS

DriveThru Voice Ordering Modernization *VoicePlug Inc.*

- Shipped production features for a real-time, multi-tenant DriveThru voice ordering agent built on FastAPI + Socket.IO with a LangGraph (ReAct) agent over Google Gemini 2.5-flash, with automatic fallback to Gemini 2.0-flash and per-call token-cost accounting.
- Refactored entity parsing, size normalization, and combo/multi-size guard logic to eliminate cart crashes and enforce semantically valid orders (e.g., wings overrides, mandatory “with Fries” confirmation) before persistence.
- Hardened entity extraction against ASR disfluencies (partial names, fillers, mispronunciations) using fuzzy matching and added latency-masking filler scheduling that suppresses duplicate TTS when the real model response arrives in time.
- Worked across 4 POS integrations (QUBeyond, Clover, Speedline, Deliverect) and Jinja2-templated per-restaurant prompts, with Redis-backed session state and MongoDB-backed order-event audit trail.
- Helped lift live client success metrics from ~25–30% to 60%+ within months by tightening intent recognition, ambiguity handling, and fallback/escalation behaviour.

HalalLens – Halal Investment Research Copilot *Open-Source / Personal Project*

- Designed and shipped a full-stack Shariah-compliance research platform – FastAPI (33 endpoints), Next.js 14 + TypeScript, PostgreSQL, Docker, GitHub Actions CI.
- Built an AAOIFI-style ratio engine across 5 compliance screens with a dual-mode “Explain Why” narrator (rule-based + vLLM / OpenAI-compatible LLM, citation-grounded).
- Engineered the data pipeline: BSE + NSE announcement crawlers, PDF/XBRL extraction (camelot, lxml) into PostgreSQL, polling workers re-evaluating compliance verdicts on a timer.
- Wired Google OAuth + JWT, watchlists with SMTP alerts on compliance flips, and Razorpay payment endpoints (order / verify / webhook) scaffolded for sandbox integration.

G6 Voice Assistant – Natural Language to SQL Converter *Global InfoVentures Pvt. Ltd.*

- Integrated SpeechRecognition, Rasa, and Gemini 2.0 to convert user voice input into structured SQL queries, enabling seamless access to ERP data from modules like Admission, Employee, Fee, and Library
- Achieved ~100% intent classification accuracy using Rasa by training with 50+ voice query variations per module. Devised a secondary intent resolution via contextual follow-ups for ambiguous queries (example, "Hamza Aziz" as student or employee)
- Formulated 100+ field-level schema mappings per module for Gemini and integrated Milvus Vector DB-based suggestions, reducing LLM hallucinations and boosting SQL query precision by over 30%

- Crafted a voice-adaptive summary module that transformed 1000+ SQL outputs into natural language, enhancing end-user experience in production deployment
- Solved heterogeneity in field values (example, department / designation name variations) using vector-based field and value disambiguation, improving precision in SQL generation by over 40%

CCTV-Based Face Attendance System *Global InfoVentures Pvt. Ltd.*

- Launched on NVIDIA DGX A100, orchestrated 20+ live CCTV streams using `scrfd_10g_gnkps` and `w600k_r50` models (TensorRT Optimized)
- Pioneered dual dataset staging (mobile & CCTV) with Milvus Vector DB to mitigate domain shift, achieving over 99.99% face recognition accuracy across varied environments
- Architected a CI/CD-style data pipeline that extracts facial data from elevated-angle surveillance cameras, filters best-quality faces per person, and auto-updates vector database, improving recognition rates by 25% in practical deployment scenarios

PTZ-Classroom Attendance System *Global InfoVentures Pvt. Ltd.*

- Automated attendance tracking by deploying rotating PTZ cameras and integrating with face recognition pipelines, eliminating manual roll calls and reducing proxy attendance by 25%

Edge-Based Office Attendance System *Global InfoVentures Pvt. Ltd.*

- Commissioned Jetson Orin Nano for real-time edge inference using DeepStream SDK, Raspberry Pi AI camera, and proximity sensors to enable on-device facial recognition
- Eliminated manual attendance logging and replaced legacy biometric systems with 100% contactless AI-driven tracking — enhanced hygiene and operational efficiency in office-based setups

CERTIFICATES

- LangChain – Develop AI Agents with LangChain & LangGraph | Udemy - 2025
- Generative AI with Large Language Models | Coursera - 2024
- Building Video AI Applications at the Edge on Jetson Nano | NVIDIA Deep Learning Institute - 2024
- Getting Started with AI on Jetson Nano | NVIDIA Deep Learning Institute - 2024
- Artificial Intelligence and Machine Learning | Global InfoVentures Pvt. Ltd. - 2023
- Getting Started with Deep Learning | NVIDIA Deep Learning Institute – 2022